NEO / GICForge Manual

Non-redundant generalized internal coordinates and symmetry-adapted coordinate bases in MATRIX

ORACLE Development Notes

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Abstract

NEO, currently exposed through the compatible `GICForge` command name, is the `ORACLE` geometry backend that converts Cartesian molecular structures into non-redundant generalized internal coordinates (GICs), optional symmetry-adapted GICs, and symmetry-adapted Cartesian coordinates. It is used as a library service by MORPHEUS, SEfit and future geometry optimizers, and can also be run as a standalone diagnostic tool. This manual describes the input contract, coordinate-generation policy, pruning and symmetrization rules, output files, Python/Fortran identity checks and recommended regression tests.

Contents

1 Purpose

NEO/GICForge has one responsibility: starting from a Cartesian molecular structure, it constructs a chemically interpretable, rank-complete coordinate basis. The basis can be used directly as a Gaussian ReadGIC block, frozen as a machine-readable `ORACLE` schema, evaluated analytically on later Cartesian geometries, or converted into symmetry-adapted Cartesian displacements.

`GICForge` deliberately does not own project management, SMILES handling, electronic-structure parsing, DVR, semiexperimental least-squares logic or GUI workflow orchestration. Those tasks belong to the Python layer. The geometry backend owns topology perception from Cartesian coordinates, primitive internal coordinates, ring handling, type-local redundancy removal, symmetry labels and the final auditable coordinate definition.

2 Theoretical Background

Let x be the $3N$-dimensional Cartesian coordinate vector and let $q_p(x)$ be a primitive internal coordinate. `GICForge` first builds a redundant primitive set

$$\mathbf{q}(x) = \{q_1(x), q_2(x), \dots, q_M(x)\},$$

where M is usually larger than the vibrational rank. A generalized internal coordinate is a linear combination of primitives,

$$Q_k(x) = \sum_p U_{pk} q_p(x),$$

with U stored in the frozen GIC definition. The Wilson row associated with the coordinate is

$$\mathbf{B}_k = \frac{\partial Q_k}{\partial x} = \sum_p U_{pk} \frac{\partial q_p}{\partial x}.$$

The final coordinate set is valid only if the B matrix has rank $3N - 6$ for a nonlinear molecule or $3N - 5$ for a linear molecule.

The central algorithmic choice is that redundancy removal is performed by coordinate family, not by a global decomposition over all candidates. In a global SVD or global rank-revealing decomposition, the numerical ordering of rows can make chemically unrelated coordinates compete with one another. A stretch row could remove a torsional row, or an out-of-plane row could remove a ring bend, even though the two coordinates describe different physical motions. `GICForge` therefore uses the already accepted rows as a rank basis and admits additional rows only within a fixed family order. Inside each family, a modified Gram–Schmidt rank test keeps the rows that add linearly independent B-matrix information. This preserves interpretability while still enforcing the exact vibrational rank.

Symmetry adaptation is a second projection problem. Once the final post-pruning basis is known, equivalent coordinate rows are projected with the molecular symmetry operations and labelled by irreducible representation. The projection is constrained to homogeneous coordinate blocks, so the symmetry operation cannot manufacture a coordinate by mixing unrelated physical families. The theoretical target counts are those of the vibrational representation,

$$\Gamma_{\text{vib}} = \Gamma_{3N} - \Gamma_{\text{trans}} - \Gamma_{\text{rot}}.$$

If the projected coordinate counts cannot reproduce this representation, the coordinate generator fails rather than writing an ambiguous basis.

3 Input Contract

The molecular geometry input is the XYZ-format file `xyzin` in the working directory. Atomic symbols and atomic numbers must be equivalent and must cover the full periodic table through the shared `ORACLE` periodic-table utilities. The `provin` file contains only keywords, title, charge and multiplicity; it must not contain Cartesian coordinates, Z-matrices or FCHK-derived geometry.

```
# GNIC BMAT GICSYM

Pyrene GICForge example

0 1
```

Unsupported legacy geometry paths are intentionally disabled: Z-matrix input, Cartesian coordinates embedded in `provin`, FCHK geometry input, SMILES input, and old FITPOT/VCI/DVR/MSR/isotope/rate driver keywords. Cartesian input is the only production path.

4 Build and Execution

The active Fortran backend is built with:

```
cd engines/fortran/gicforge
./compile_legacy
```

The build writes `engines/fortran/gicforge/build/gicforge_legacy`.

The recommended command-line entry point is the shared `ORACLE` CLI:

```
python -m oracle gic --workdir work/gicforge
python -m oracle gic-define geometry.xyz --workdir work/gicdefine
python -m oracle gic-bmatrix oracle.gic.definition.json geometry.xyz
python -m oracle gic-contract --geometry geometry.xyz --workdir work/contract
```

`gic-define` creates a frozen schema; `gic-bmatrix` evaluates that schema without rerunning topology perception; `gic-contract` verifies the Fortran/Python identity contract.

5 Coordinate Construction Algorithm

The coordinate generator is a deterministic pipeline. Each step consumes only information created by earlier steps, and the final schema records enough metadata to verify that the same model is reused later.

5.1 Topology perception

`GICForge` starts from interatomic distances and atomic covalent radii. The result is a covalent graph with atoms as vertices and bonds as edges. The graph is checked before coordinate construction: symbols and atomic numbers must be consistent, atom indexes must be unique, and obvious nonbonded H–H contacts or other impossible short contacts are treated as input/topology errors. The graph is then frozen for the run. Later refinement or optimization modules must not allow a geometry step to change this graph silently.

Ring perception is performed on the covalent graph. The ring basis is canonicalized before any ring coordinate is built, so the same molecule gives the same ring order in Fortran and Python.

Fused systems are represented by their individual rings plus shared bonds; the shared bonds are the places where butterfly coordinates can be introduced.

5.2 Primitive coordinate generation

The first coordinate layer is primitive and usually redundant. The generator adds primitives in chemically ordered families:

1. stretches for every covalent edge;
2. ordinary valence angles around each bonded center;
3. linear-bend candidates only for nearly linear centers, with a maximum number per center;
4. exocyclic torsions from the default one-dihedral rule;
5. endocyclic valence-angle primitives for each ring;
6. butterfly primitives around bonds shared by two rings;
7. endocyclic puckering components `RPck` for each ring;
8. out-of-plane primitives, except when all four involved atoms are cyclic;
9. optional improper-dihedral primitives in the Gaussian-compatibility mode.

This order is not only cosmetic. It defines the chemical priority used by the rank reduction. Stretches define the bonded frame first; bends define local coordination; torsions and ring coordinates define conformational and puckering directions; out-of-plane coordinates close the remaining local non-planarity directions.

5.3 Local combinations before pruning

Several local coordination patterns are better represented by combinations than by a single primitive. For example, equivalent valence angles around a highly symmetric center may be converted to local sums and differences, and ring torsions are converted to deformation coordinates. These combinations are still type-local: angles are combined with angles, torsions with torsions, and out-of-plane coordinates with out-of-plane coordinates. This rule is essential because it prevents the construction stage from creating mathematically valid but chemically opaque coordinates.

For ring systems, endocyclic torsions are not meant to appear as an arbitrary list of raw dihedrals in the final model. They are the primitive source from which `GICForge` builds the active q- and ϕ-like ring deformation coordinates. ORACLE stores the linear `RPck` components in the frozen GIC definition because their B rows are analytic linear combinations of dihedral B rows. The Gaussian block writes those `RPck` components as inactive auxiliaries and derives the selected active `QPck` and `PhiP` functionals from consecutive `RPck` pairs, matching Merlino3.0 `PrtPckQP`.

5.4 Rank reduction and final basis

After primitive generation and local combinations, `GICForge` evaluates Wilson B rows for the candidate coordinates at the current Cartesian geometry. It then keeps rows in the fixed family order and rejects rows that do not increase the rank. The rank test is numerical, but the candidate competition is chemical: an angle can remove another redundant angle, but not a stretch; a ring bend can be removed by ring-bend redundancy, but not by an unrelated exocyclic torsion.

If the ordinary candidate set is too small before pruning, the backend expands within Fortran to the full primitive candidate set and repeats the same type-local pruning. The number of pre-pruning attempts is capped by the number of available primitive coordinates. If the final count is still not $3N-6$ or $3N-5$, the run stops. A downstream solver must never repair the basis by inventing an independent coordinate model.

6 Coordinate Definition Format

Coordinate definitions are printed in a Gaussian ReadGIC-compatible form. Each line has a name on the left-hand side, an equal sign, and either one primitive or a normalized linear combination of primitives on the right-hand side:

```
Str0001 = R(1,2)
Bend0007 = A(2,1,6)
Tors0014 = D(3,4,5,6)
OuPl0003 = U(7,2,1,3)
GIC0021 =  0.70710678 A(2,1,6) -0.70710678 A(3,4,5)
```

Atom indexes are one-based and always refer to the input atom order after any explicit canonical ring ordering has been recorded in the report. The primitive symbols are:

Primitive	Meaning
`R(i,j)`	Bond stretch between atoms i and j.
`A(i,j,k)`	Valence angle $i-j-k$, with j as the central atom.
`L(i,j,k)`	Linear-bend component for a nearly linear $i-j-k$ arrangement.
`D(i,j,k,l)`	Dihedral angle around the central bond $j-k$.
`U(i,j,k,l)`	Out-of-plane coordinate with a single central atom. The Fortran and Python parsers enforce the same central-atom convention.
`ImpD(i,j,k,l)`	Improper-dihedral compatibility form used only with `G16` or `ImpDih`.

The name prefix indicates the coordinate family and is part of the stable ordering contract:

Prefix	Family
`Str`	primitive stretch;
`Bend`	exocyclic or final valence-angle coordinate;
`LinB`	linear-bend coordinate;
`Tors`	exocyclic torsion;
`BtFl`	butterfly torsion around a bond shared by two rings;
`CyGNA`	endocyclic valence-angle combination;
`QPck`	active ring puckering amplitude-like coordinate built from endocyclic torsions;
`PhiP`	active ring puckering phase-like coordinate built from endocyclic torsions;

Prefix	Family
`RPck`	inactive Gaussian auxiliary and frozen ORACLE linear ring-puckering component with analytic B row;
`OuPl`	out-of-plane coordinate;
`ImpD`	improper-dihedral compatibility coordinate.

6.1 Ring Coordinate Format

For a ring with ordered atoms $a_1, a_2, \ldots, a_n$, `GICForge` generates two ring families. Endocyclic valence-angle coordinates are combinations of primitive angles

$$A(a_{i-1}, a_i, a_{i+1}),$$

where indexes are cyclic. Endocyclic torsional coordinates are combinations of primitive dihedrals

$$D(a_{i-1}, a_i, a_{i+1}, a_{i+2}).$$

The primitive torsions are first transformed to `RPck` components. The frozen ORACLE B matrix uses these linear components directly. For Gaussian ReadGIC export, consecutive selected `RPck` pairs are written as inactive auxiliaries and converted to active puckering coordinates `QPck` and `PhiP`:

$$Q = \sqrt{RPck_i^2 + RPck_j^2}, \qquad \Phi = \operatorname{atan2}(RPck_j, RPck_i).$$

For a five-membered ring, the report therefore separates the coordinate construction from the active coordinates:

```
Endocyclic Dihedral Angles
  5-Membered Ring (  13  12  16   1   2):  2 Dihedral angles

Final ring torsion coordinates
  RPck0001(Inactive) = ...
  RPck0002(Inactive) = ...
  QPck0001 = ...
  PhiP0001 = ...
```

For six-membered and larger rings, the same rule is used: the primitive basis is endocyclic torsional, but the Gaussian-active coordinates are the independent q- and ϕ-like ring deformation coordinates derived from selected `RPck` pairs. If an odd component remains after pairing, it stays as an inactive `RPck` auxiliary, as in Merlino3.0. Fused rings add `BtFl` butterfly coordinates around shared bonds. Butterfly coordinates remain in the torsional family for pruning because their B rows describe ring-puckering motion.

7 Coordinate Families

`GICForge` builds primitive coordinates from the covalent graph. Hydrogen bonds may be detected and reported by higher-level topology utilities, but the standard GIC graph remains covalent: hydrogen bonds are correction targets, not ring-closing covalent edges.

Family	Policy
Stretch	All bonded pairs are kept as primitive `R(i,j)` coordinates. Stretches are not reduced by the final pruning step.
Exocyclic bend	Valence angles outside rings are generated around each center and reduced locally by rank.
Linear bend	Linear-angle candidates are generated only when required by local geometry and are capped per center to avoid unphysical proliferation.
Exocyclic torsion	The default torsion generator is the one-dihedral `ONEDIH` path. The keyword `NOONEDIH` forces the older alternative for diagnostics.
Ring bend	Endocyclic valence-angle combinations are generated ring by ring, with canonical ring numbering.
Ring torsion	Endocyclic torsions are converted to `RPck` components before pruning. The frozen ORACLE basis keeps selected `RPck` rows for linear analytic B-matrix evaluation; the Gaussian block marks them inactive and adds active `QPck`/`PhiP` functionals for complete pairs.
Butterfly	Fused-ring butterfly coordinates are generated around shared bonds and kept as torsional ring-family coordinates.
Out-of-plane	Default non-planarity primitives are `U(...)` out-of-plane angles. They are not generated when all four involved atoms are cyclic. Cyclic centers with exocyclic substituents remain eligible.
Improper dihedral	With `G16` or `ImpDih`, improper dihedrals are written instead of out-of-plane angles. This is a compatibility mode and must match the Python parser exactly.

8 Rings and Polycyclic Systems

Ring atom ordering is canonicalized with the same Prelog-first convention used by the Python implementation. The `provout` report lists atoms in one cycle, atoms linking multiple cycles, rings found versus expected rings, endocyclic valence-angle coordinates, endocyclic torsional coordinates, butterfly coordinates and out-of-plane candidates.

For planar polycyclic aromatic hydrocarbons, out-of-plane candidates whose four atoms are all cyclic are suppressed. Exocyclic out-of-plane coordinates remain available and are either kept or removed by the final rank test. Regression targets should include naphthalene, anthracene, pyrene and coronene, with explicit checks that fused-ring butterfly coordinates are present when topologically required.

9 Redundancy Removal

The final pruning step is type-local. `GICForge` builds B rows for candidate GICs and applies a modified Gram–Schmidt rank test separately to homogeneous coordinate families:

- exocyclic bends;
- linear bends;
- exocyclic torsions;

- butterfly and ring torsional coordinates;
- ring valence-angle coordinates;
- out-of-plane coordinates.

Stretch rows are always retained as primitive coordinates. This policy avoids the chemically undesirable behavior of a global SVD in which, for example, a stretch can eliminate a torsion simply because it appears earlier in the global rank basis. The rank test is still numerical and B-matrix based, but the admissible competition is restricted to coordinates with the same physical role.

Fragment and virtual-center coordinates are treated as a separate protected class. These coordinates include fragment-center distances, fragment-center to atom distances, fragment translations, fragment orientations and atom to virtual-center distances for ring centers, bond centers and contact centers. They are not ordinary valence-graph primitives, and they can carry chemical information that no single stretch, bend or torsion should be allowed to erase. For this reason `GICForge` first tests all protected special coordinates against each other by the same analytic B-matrix modified Gram–Schmidt rank test used by the Fortran77 routine `ORCGSEL`. Only after that protected pass does it use ordinary primitives to complete the vibrational rank. A special coordinate may be skipped only when its own row is singular or linearly dependent on previously accepted special rows. If the independent protected set alone is larger than the vibrational rank, `GICForge` stops with a contract error instead of silently deleting a protected coordinate.

The frozen `xyzin` schema records this distinction explicitly. Ordinary coordinates are written with `CLASS=ORDINARY`. Fragment and virtual-center coordinates are written with the protected class token `SPECIAL_PROTECTED`. The header also records the rank method `analytic_b_matrix_mgs_greedy`. The reduction policy is the protected-first policy described above, serialized in `xyzin` as two adjacent tokens: `SPECIAL_PROTECTED_FIRST_THEN_ORDINARY_` and `ANALYTIC_RANK`. This makes restarts reproducible and tells downstream programs that the protected coordinates are part of the molecular model, not disposable redundant candidates.

The target rank is $3N-6$ for nonlinear molecules and $3N-5$ for linear molecules. If the final non-redundant count does not match the target, `GICForge` stops with an error. If the pre-pruning count differs from the target, coordinate definitions are not printed as final coordinates until after pruning has produced the usable basis.

10 Symmetry

`GICForge` uses symmetry in three different places, and these roles should not be confused.

10.1 Point-group perception

The first role is molecular point-group perception. `GICForge` links the shared Fortran symmetry engine `symm.f`. Cartesian coordinates are centered at the center of mass, oriented on principal inertia axes and assigned a point group. The authoritative report is the point group computed by `symm.f`; an optional point-group token on the second `xyzin` line is legacy metadata only.

The symmetry report records the detected point group, the quality of the assignment and the maximum atom-matching deviation. A high-symmetry assignment is used only when atom permutations and Cartesian operations are consistent within the configured tolerances. If the geometry is only approximately symmetric, the report must make that explicit, because the symmetry-adapted coordinate basis is only as meaningful as the atom mapping used to build it.

10.2 Symmetry labels for final GICs

The second role is assigning irreducible representations to the final post-pruning GIC basis. This is done after the non-redundant coordinate set has been selected. The order matters: symmetry is not allowed to rescue a redundant coordinate set, and pruning is not allowed to destroy the final irreducible-representation counts. The target representation is

$$\Gamma_{\text{vib}} = \Gamma_{3N} - \Gamma_{\text{trans}} - \Gamma_{\text{rot}}.$$

For a nonlinear molecule, the sum of the target irrep counts is therefore $3N - 6$.

With `GICSYM`, the deterministic Python post-check projects the final GIC rows onto symmetry-adapted combinations, assigns irreducible representations, writes the symmetry-adapted GIC block and records the result in:

- `gauin.symm`;
- `gicsym`;
- `gic_symmetry_diagnostics.json`.

The promoted `gauin` file contains the symmetrized Gaussian block. Totally symmetric coordinates are written first, followed by a blank separator and then the other irreducible representations. Downstream programs must not re-symmetrize; they must consume the frozen schema and its labels.

10.3 Allowed symmetry blocks

The post-check is deliberately constrained. A symmetry-adapted coordinate must be expressible from a physically homogeneous source block. The allowed blocks are primitive stretches, ordinary bends, linear bends, torsions, out-of-plane coordinates, ring bends, ring torsions, local out-of-plane/improper blocks around the same center and protected special-coordinate blocks. Protected special-coordinate blocks are fragment-center distances, fragment-center to atom distances, fragment translations, fragment orientations and atom to virtual-center distances. There is no global fallback that mixes every primitive in the molecule. This preserves the meaning of the final coordinates and makes the result comparable between Fortran and Python.

Within a homogeneous block, equivalent primitives are collected by the symmetry operations. The totally symmetric combination is the normalized sum over the orbit. Non-totally-symmetric combinations are orthonormal differences or the corresponding projection-operator rows. The sign is canonicalized so repeated runs produce the same labels and coefficients. The post-check then verifies both irrep counts and coordinate-family counts. For example, a missing torsional B_{2g} coordinate cannot be replaced by an extra stretch of the same irrep.

10.4 Fortran local SALCs versus production symmetry

The keyword `SYMMALL` activates a limited Fortran-side local SALC pass for selected non-stretch one-term coordinates. This is a construction aid, not the owner of production symmetry. Production symmetry adaptation is the `GICSYM` post-check, because it sees the final pruned basis, can compare against the theoretical vibrational representation and writes the frozen schema. Stretches remain primitive in the raw `gauin` and are symmetry-adapted only in the post-check when `GICSYM` is requested.

10.5 Symmetry-adapted Cartesians

With `SYCART`, `GICForge` writes symmetry-adapted Cartesian coordinates after removal of rotations and translations. This provides a common service for SEfit, MORPHEUS and future optimizers without embedding symmetry logic in those modules. The Cartesian basis follows the same point-group assignment as the GIC basis. It is useful when a downstream module wants a translationally and rotationally clean totally symmetric displacement space without using GICs.

10.6 Failure policy for symmetry

Symmetry adaptation fails if the projected basis cannot reproduce the target vibrational irrep counts, if coordinate-family counts change, if atom permutations are inconsistent, or if two equivalent source blocks would produce ambiguous labels. These failures are intentional. A wrong or ambiguous symmetry label would contaminate downstream refinement, Gaussian optimization or force-field projection more seriously than a clean stop.

11 Frozen Schema and Public API

The production public API is the `GICForge` service in `oracle_gic`. A fresh definition run writes a `oracle.gic.definition.v1` schema containing:

- primitive coordinate expressions;
- GIC coefficient matrix;
- coordinate labels and names;
- coordinate families;
- irreducible representations;
- point group and symmetrization flag;
- Gaussian-readable GIC block;
- provenance hashes for input, backend and generated files.

The schema is frozen during an ordinary refinement or optimization. Later iterations evaluate values and B rows from the schema without changing topology, ring detection, redundancy removal or symmetry assignment. A restart is the explicit boundary at which the schema may be regenerated.

12 Output Files

File	Meaning
`provout`	Human-readable report: point group, topology summary, candidate families, pruning diagnostics, final counts and final coordinate definitions.

File	Meaning
`gicforge_report.txt`	ORACLE-native report generated from frozen `xyzin` state with rank method, reduction diagnostics, family counts, protected-coordinate counts, symmetry source blocks and final GIC labels.
`gauin`	Gaussian ReadGIC input block for the final post-pruning coordinate basis. With `GICSYM`, this is the promoted symmetrized block.
`gauin.raw`	Preserved unsymmetrized Gaussian block when `GICSYM` promotes `gauin.symm`.
`gauin.symm`	Symmetry-adapted Gaussian block before promotion.
`gicsym`	Names, irreps and source-block information for the symmetrized coordinates.
`bmat.out`	Machine-readable final B matrix when `BMAT` is active.
`bondout`	Detected covalent topology.
`ringin`	Canonical ring data used by the ring-coordinate generator.
`sycart.xyz`	Symmetry-adapted Cartesian coordinates when `SYCART` is active.
`oracle.gic.definition.v1.json`	Frozen reusable GIC schema written by `gic-define`.

13 Keyword Summary

Keyword	Effect
`GNIC`	Build generalized natural/internal coordinates.
`BMAT`	Write the machine-readable final B matrix.
`GICSYM`	Produce symmetry-adapted GICs and promote them to the Gaussian input block.
`SYCART`	Produce symmetry-adapted Cartesian coordinates without rotations and translations.
`GDV`	Use out-of-plane `U(...)` coordinates; this is also the default behavior.
`G16`, `ImpDih`	Use Gaussian-style improper dihedrals instead of out-of-plane coordinates.
`ONEDIH`	One-dihedral torsion generation. This is the default.
`NOONEDIH`	Force the legacy alternative torsion path for diagnostics.
`SYMMALL`	Apply the limited type-local Fortran SALC generator before final pruning. Production symmetry adaptation is still owned by `GICSYM`.

14 Python/Fortran Identity Contract

The Python layer is allowed to orchestrate, serialize, post-check symmetry and evaluate the frozen B matrix. It is not allowed to replace a valid Fortran GIC basis silently. The contract checks:

- final coordinate count and family counts;

- ordered primitive signatures;
- point group and irreducible-representation labels;
- number of totally symmetric coordinates;
- Python analytic B matrix against Fortran `bmat.out`;
- byte-stable symmetry diagnostics for repeated runs.

Run the contract with:

```
python -m oracle gic-contract --geometry examples/pyrene.xyz \
  --workdir work/gic_contract_pyrene
```

The imported Merlino3.0 strict Fortran77 backend is vendored and compiled with:

```
engines/fortran/gicforge/legacy_merlino
engines/fortran/gicforge/compile_legacy
```

ORACLE-specific Fortran fragment/TRIC kernels remain in:

```
engines/fortran/gicforge/frag_tric_bmat.f
```

Both layers must mirror `oracle_gicforge.policy`: primitive families, reduction classes, rank method, protected-first pruning and symmetry source blocks are shared contract data, not backend-local choices.

The readable ORACLE report is generated directly from the frozen `xyzin`:

```
oracle gicforge report molecule.xyzin gicforge_report.txt
```

The Python port in `oracle_gic.gicforge_python` is a faithful porting target, not an independent production generator. It must match Fortran on ordering, primitive orientation, coordinate families, type-local pruning, symmetry labels and B rows before it can replace a backend path.

15 Recommended Regression Set

The minimum regression set should cover:

- acyclic molecules with ordinary stretches, bends and torsions;
- linear-angle cases with capped linear bends;
- square-planar or high-coordination centers;
- SF_6, which should not produce spurious linear bends;
- saccharine and other heteroatom rings;
- naphthalene, anthracene, pyrene and coronene for fused PAHs;
- norbornene and norbornadiene for bridged non-planar rings;
- `GICSYM` variants for nontrivial point groups;
- `SYCART` variants for symmetry-adapted Cartesian output.

Every test should verify the target rank, coordinate-family counts, presence of required butterfly coordinates, absence of all-cyclic out-of-plane primitives, and equality of Python/Fortran B matrices where both paths are active.

16 Failure Policy

`GICForge` must fail early when the generated basis is not scientifically usable. Fatal conditions include:

- inconsistent symbol/atomic-number input;
- topology changes during a frozen-schema workflow;
- final GIC count different from $3N - 6$ or $3N - 5$;
- symmetry post-check unable to reach theoretical irrep counts;
- coordinate-family counts changed by post-check symmetrization;
- Python/Fortran B-matrix mismatch beyond numerical roundoff.

These failures are intentional. A downstream fit or optimizer must not proceed with an ambiguous coordinate definition.

17 Relation to MORPHEUS and SEfit

MORPHEUS and SEfit call `GICForge` as a coordinate-service library. A fresh fit generates one frozen coordinate definition at the starting geometry. Ordinary least-squares iterations then reuse that definition and only update coordinate values, B rows and projected parameter directions. This prevents the fitted model from changing during optimization and makes the final refinement auditable.

For a GIC active model, the fit uses the totally symmetric block assigned by `GICForge`. For a symmetry-adapted Cartesian active model, the fit uses `SYCART`. In both cases, topology and symmetry belong to `GICForge`, while weights, constraints, predicates, parameter classes and least-squares diagnostics belong to the refinement layer.